

Test Plan

Selected Project: School Fee Payment Tracking System

Team Members:

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Student.py

Test ID	Method / Feature	What we are testing	Input	Expected Result
TC-S01	Register student	Check that we can register a student when all the information entered is correct.	ID: S010, Name: Alice Smith, Grade: Grade 10, Fee: 500	The student should be registered successfully and added to the student list.
TC-S02	Display Student Details	Check that the system displays the correct information for a registered student.	S010, Alice Smith, Grade 10, Fee: 5000	The student's ID, name, grade and total fee should be displayed correctly.
TC-S03	Student ID Validation	Check what happens when the user does not enter a Student ID.	Student ID: empty	The system should show an error message and should not register the student
TC-S04	Student ID Validation	Check that the system does not	Student ID: S010, when	The system should show an

		allow two students to have the same Student ID	S010 already exists	error message saying that the Student ID is already being used.
TC-S05	Register student	Check what happens when a negative fee is entered	Fee: -5000	The system should show an error message and should not register the student.

Payment.py

Test ID	Method / Feature	What we are testing	Input	Expected Result
TC-P01	Record Payment	Check that a normal payment can be recorded successfully	Outstanding balance: 5000, Payment: 2000	The payment should be accepted and recorded successfully.
TC-P02	Payment Amount Validation	Check what happens when the user enters zero as the payment amount.	Payment: 0	The system should reject the payment and show an error saying the payment must be greater than zero.

TC-P03	Payment Amount Validation	Check what happens when the user enters a negative payment.	Payment: -100	The system should reject the payment and show an error message.
TC-P04	Outstanding Balance Check	Check that a student cannot pay more than what they owe.	Outstanding balance: 3000, Payment: 4000	The system should reject the payment and show an error message.
TC-P05	Outstanding Balance Check	Check what happens when the student pays exactly the amount they still owe.	Outstanding balance: 3000, Payment: 3000	The payment should be accepted and the outstanding balance should become \$0.00

FeeAccount.py

Test ID	Method / Feature	What we are testing	Input	Expected Result
TC-F01	Calculate Total Paid	Check that the system correctly adds several payments together.	Payments: 1000, 1500, 500	The total amount paid should be 3000.
TC-F02	Calculate Outstanding Balance	Check that the system correctly calculates what the student still owes	Total fee: 5000, Total paid: 3000	The outstanding balance should be 2000.

TC-F03	Add Payment	Check that a payment is added when it belongs to the correct student.	Account ID: S010, Payment ID: S010	The payment should be added to the student's payment history
TC-F04	Add Payment	Check what happens when a payment belongs to a different student.	Account ID: S010, Payment Student ID: S011	The payment should not be added and an error message should be displayed.
TC-F05	Display Payment History	Check what the system displays when a student has not made any payments.	Student with no payments	The system should say that no payments have been made and display the total fee, total paid and outstanding balance.

Test Case Code Meaning

TC = Test Case

S = Student

P = Payment

F = FeeAccount

01, 02, 03... = Test case number

Examples:

- **TC-S01** = Test Case – Student – Test 01
- **TC-P01** = Test Case – Payment – Test 01
- **TC-F01** = Test Case – FeeAccount – Test 01

